

RAYENE MEDJTOH

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OBJECTIVE

Software engineering internship, July to September 2027. Relocating for the duration of the internship.

EDUCATION

CESI Engineering School Sept. 2023 – Sept. 2028 (expected)
M.Eng. in Computer Engineering — 5-year integrated program (master's level) Aix-en-Provence, France

- Coursework: operations research, metaheuristics, algorithms and complexity, distributed systems, databases, network architecture

EXPERIENCE

Software Engineer Oct. 2025 – Present
Sopra Steria — work-study program (alternating full-time work and coursework) Aix-en-Provence, France

- Build and maintain Java components on a 10-model TIBCO EBX Master Data Management platform for Arkema, a global specialty chemicals group, within a 4-person Agile team
- Developed and debugged 20+ data-governance workflows, including a premature-termination defect that was dropping user submissions before they reached validation
- Automated anonymization of confidential records across 600+ tables, replacing a manual record-by-record process that bottlenecked every test-environment setup
- Root-caused and fixed 60+ production defects reported by client business users, from array-bounds failures to node-resolution errors in the workflow engine

Software Development Intern Apr. 2025 – July 2025
ArcelorMittal Fos-sur-Mer, France

- Developed a configurable SQL Server monitoring service in C# (.NET), letting site IT define watch targets and alert thresholds without changing code
- Migrated the real-time supervision interface for steelworks water discharge (flow, pH, temperature) onto the plant's certified IoT sensor database
- Repointed ASURE, the application producing the plant's monthly environmental filings to the regulator, to that same certified source; deployed Prometheus across environmental servers and workstations

PROJECTS

Vehicle Routing with Time Windows | *Python, metaheuristics, deep learning* 2026

- Implemented and compared four VRPTW solvers: a nearest-neighbor heuristic as baseline, simulated annealing, a genetic algorithm, and a learned policy
- Benchmarked solution quality against runtime across instance sizes to determine which method applies at which scale, and validated result stability over repeated runs

Breezy — Microservices Social Platform | *Node.js, Express, MongoDB, Docker, NGINX* 2026

- Architected a three-service backend (authentication, users, posts) behind an NGINX gateway, orchestrated with Docker Compose, with stateless JWT authentication shared across services

Virtualized Corporate Network Infrastructure | *pfSense, Windows Server 2022, Ubuntu, DNS/DHCP* 2025

- Designed and deployed a segmented enterprise network in virtual machines: pfSense firewall/router, a Windows Server 2022 domain controller serving DNS and DHCP, an Ubuntu server, Windows 11 clients
- Configured routing, firewall rules, and name resolution end to end, diagnosing resolver and gateway errors to restore internal and outbound connectivity

TECHNICAL SKILLS

Programming languages: Java, C#, C, C++, JavaScript, Python, SQL, HTML/CSS

Frameworks: Spring, Node.js, Express, .NET, React, Redux, Mongoose

Platforms: TIBCO EBX (MDM), Docker, NGINX, SQL Server, MongoDB, PostgreSQL, Linux, Windows Server, pfSense, Prometheus

Practices: Git, GitLab CI, Maven, REST API design, Agile/Scrum

Spoken languages: French (native), English (professional working proficiency, TOEIC 800), Arabic (conversational)

Interests: competitive amateur boxing